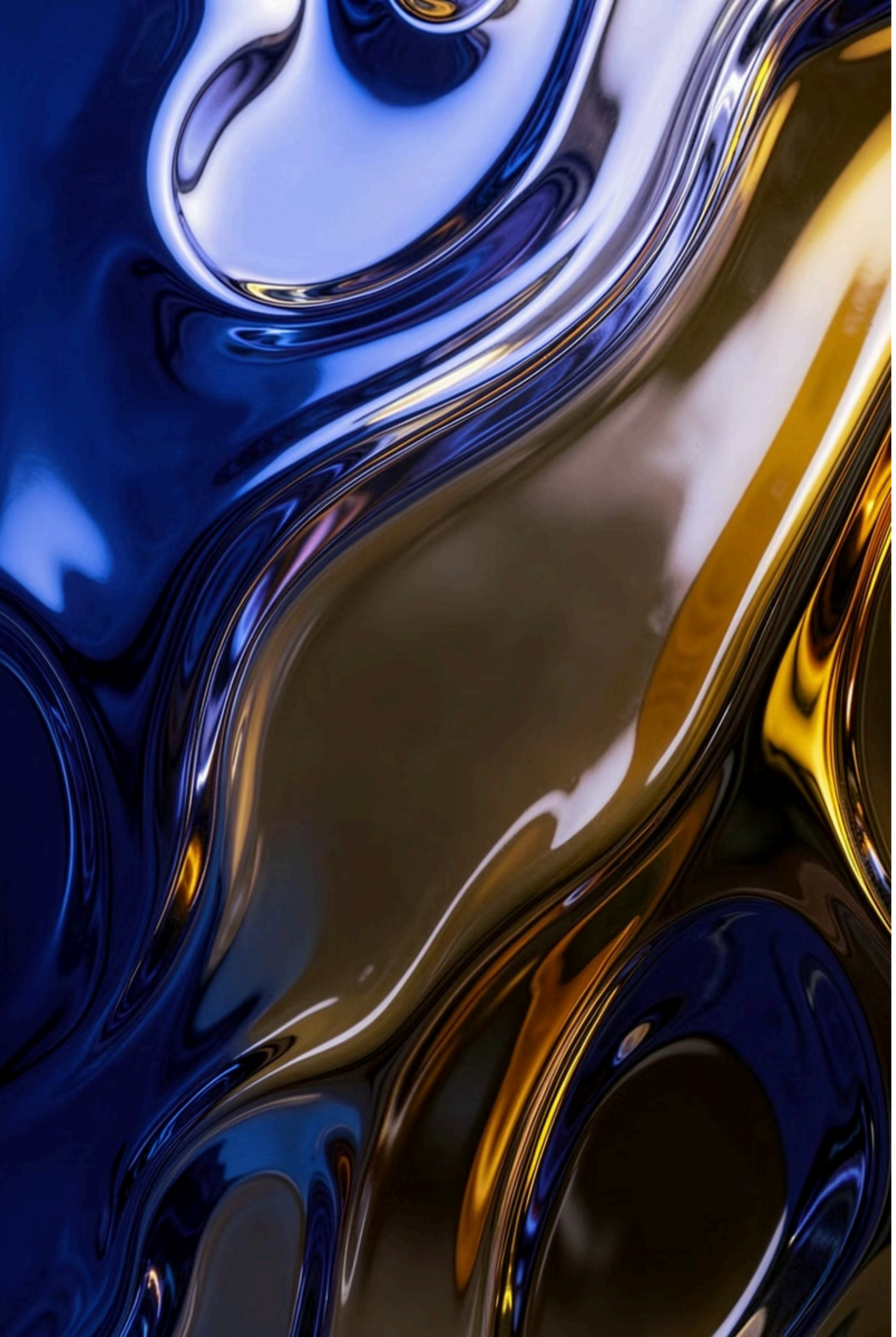


Transfer Attacks in Face Recognition

GRA — Gradient Relevance Attack

Adapted from "Boosting Adversarial Transferability via Gradient Relevance Attack" — ICCV 2023

ICCV 2023 | WHITE-BOX | GRADIENT-BASED | L_∞-BOUNDED

Presenter: Krish Bansal (DTU) | **Date:** June 2026

Executive Summary

31.46%

Overall Verified Breach Rate

Across all 480 evaluation cases

+2.29pp

Gain vs. SI-NI-FGSM Baseline

Beats the strong 29.17% baseline

40.00%

Dodging Breach Rate

Successfully evades identity matching

22.92%

Impersonation Breach Rate

Forces false cross-identity matches

GRA ranks 3rd overall across all intern attacks. Its core strength lies in neighborhood gradient smoothing and a decay indicator that prevents model-specific overfitting. The massive variance between surrogates (Facenet512 at 43.33% vs GhostFaceNet at 13.33%) reveals that surrogate choice is the single most critical factor in transfer attack success.

GRA Method Overview

ICCV 2023 | WHITE-BOX | GRADIENT-BASED | L∞-BOUNDED

DEFINITION & NATURE

- Input-transformation attack with **neighborhood gradient aggregation**
- Momentum PGD with sign-gradient updates; **ε = 0.062** (L∞), **T = 5 iterations**
- One surrogate DeepFace model per run; evaluated on **4 unseen victim models + IR152**

CORE MECHANISM

1. **The Problem** — Single-view PGD/MI-FGSM gradients overfit the surrogate model's specific architecture, destroying transferability to other models
2. **Neighborhood Sampling** — Generates 20 neighbor copies per iteration by adding bounded uniform noise, then averages their gradients
3. **Gradient Relevance** — Computes Cosine Similarity between the original gradient and the average neighborhood gradient to filter out model-specific bias
4. **Decay Indicator** — Tracks sign consistency of gradient momentum across iterations. Pixels that rapidly flip signs are penalized by γ=0.94, dampening oscillating noise

Algorithm

```
Input: source x, target embedding e_t, model f
    ε=0.062, T=5, N=20, β=3.5, γ=0.94; α←ε/T
    adv←x; g←0; m←1.0
for t = 1...T:
  // 1. Current Gradient
  L ← AttackLoss(f(adv), e_t)
  grad ← ∇_adv L
  // 2. Neighborhood Gradient Estimation
  grad_sum ← 0
  for i = 1...N:
    noise ← Uniform(-ε*β, ε*β)
    L_n ← AttackLoss(f(adv + noise), e_t)
    grad_sum ← grad_sum + ∇ L_n
  avg_grad ← grad_sum / N
  // 3. Gradient Relevance Correction
  cos_sim ← CosineSimilarity(grad, avg_grad)
  grad_corr ← cos_sim×grad + (1-cos_sim)×avg_grad
  // 4. Momentum + Decay Indicator
  prev_g ← g
  g ← g + normalize(grad_corr)
  sign_match ← (sign(prev_g) == sign(g))
  m ← m × (sign_match + (1-sign_match) × γ)
  // 5. Update
  adv ← adv + (α × m) × sign(g)
  adv ← Proj_ε(adv, x) ; clip(adv, -1, 1)
return adv
```

Evaluation Environment

Shared Perturbation Budget

Parameter	Value
ε (L∞ norm bound)	0.062
Iterations (T)	5
Momentum Decay	1.0
Neighborhood Samples (N)	20
Radius Factor (β)	3.5
Sign Decay (γ)	0.94
Step Size (α)	ε / T = 0.0124

Experimental Scale

480

Total Evaluation Cases

30 pairs × 2 goals × 4 attackers × 4 non-self victims

4

Attacker Models

Facenet512, ArcFace, GhostFaceNet, VGG-Face

5

Victim Models

Facenet512, ArcFace, GhostFaceNet, VGG-Face, IR152

2

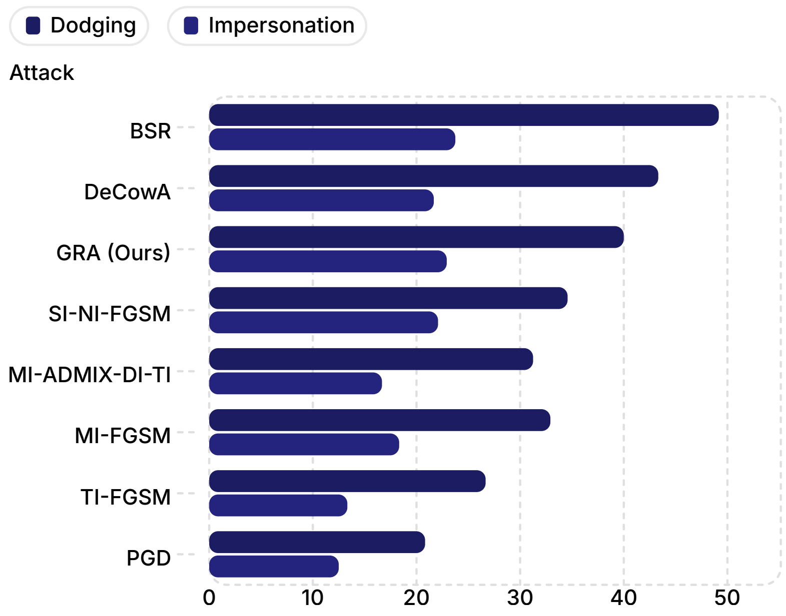
Attack Goals

Impersonation & Dodging

- **Impersonation:** fool model into matching two different identities (cosine similarity ↑)
- **Dodging:** fool model into rejecting correct identity (cosine similarity ↓)
- Self-transfer excluded from all evaluations

Overall Results vs. Baselines

Breach Rate by Attack (%)



Full Rankings

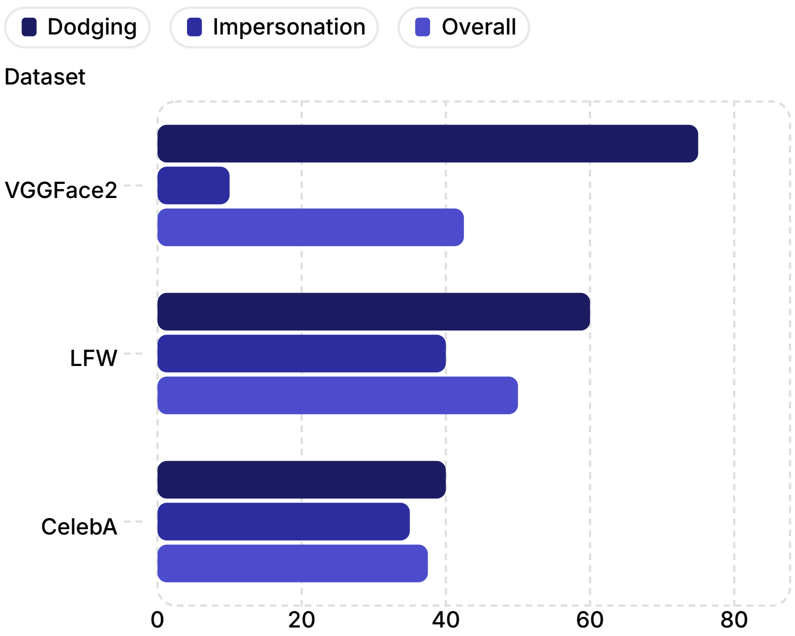
Rank	Attack	Overall
1	BSR	36.46%
2	DeCowA	32.50%
3	GRA (Ours)	31.46%
4	SI-NI-FGSM	29.17%
5	MI-FGSM	26.04%
6	MI-ADMIX-DI-TI	23.96%
7	TI-FGSM	20.00%
8	PGD	16.46%

Key Takeaways

- GRA ranks **3rd overall** at 31.46%, beating SI-NI-FGSM by +2.29pp
- GRA achieves the **highest Impersonation rate** among top 3 attacks at 22.92%
- BSR leads overall but GRA **surpasses it on Impersonation**

Performance Across Face Datasets

GRA Breach Rate by Dataset (%)



Breach Rate Breakdown

Dataset	Overall	Dodging	Impersonation
LFW	50.00%	60.00%	40.00%
VGGFace2	42.50%	75.00%	10.00%
CelebA	37.50%	40.00%	35.00%

Analysis

- LFW is GRA's strongest dataset at 50.00% — constrained single-pose images make features more predictable
- VGGFace2 Dodging is GRA's single best result at 75.00% — high-diversity images provide more gradient signal to destroy identity features
- VGGFace2 Impersonation is the hardest at only 10.00% — forcing cross-identity matches across a diverse dataset is the toughest possible task

Cross-Model Transferability Matrix

Adversarial images generated on row (attacker) → evaluated on column (victim). Diagonal (self-attacks) excluded.

Attacker → Victim Breach Rate Matrix (%)

Attacker \ Victim	Facenet512	ArcFace	GhostFaceNet	VGG-Face	IR152
Facenet512	—	46.67%	43.33%	53.33%	30.00%
VGG-Face	53.33%	50.00%	36.67%	—	10.00%
ArcFace	40.00%	—	40.00%	30.00%	16.67%
GhostFaceNet	10.00%	23.33%	—	13.33%	6.67%
Col Avg	34.44%	40.00%	40.00%	32.22%	15.84%

Row Avg = average breach rate when attacking each victim column.

Architecture Similarity Hypothesis

Transfer rates are highest between architecturally similar models. Facenet512 (Inception-ResNet) → VGG-Face (VGG-16) achieves 53.33% due to shared deep convolutional feature hierarchies. IR152 (PyTorch ResNet) is consistently the hardest victim at 15.84% avg due to its completely different training framework and backbone.

Key Insights

Best Surrogate

Facenet512 (43.33% avg) — strong convolutional features generalize well across all victims.

Worst Surrogate

GhostFaceNet (13.33% avg) — lightweight architecture generates weak, model-specific gradients.

Hardest Victim

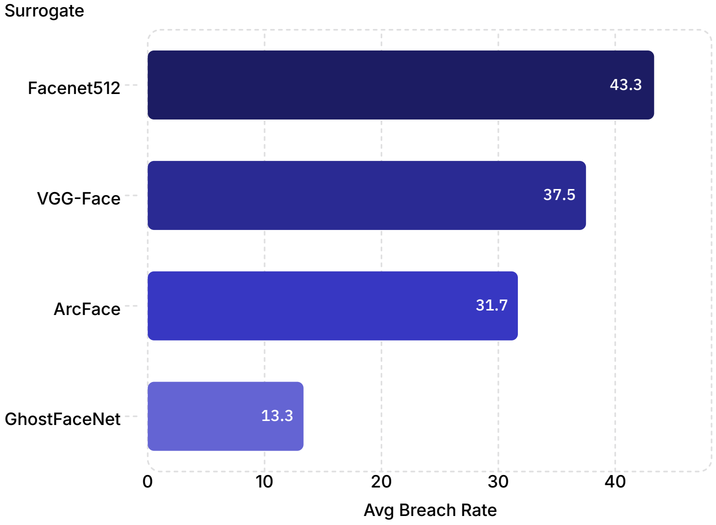
IR152 (15.84% avg) — PyTorch backbone creates a massive architecture gap from all TF attackers.

Best Pair

Facenet512↔VGG-Face (both 53.33%) — shared deep convolutional heritage.

Surrogate Model Analysis

Avg Breach Rate by Surrogate Model



Surrogate	Avg Rate	Architecture
Facenet512	**43.33%**	Inception-ResNet-V1
VGG-Face	37.50%	VGG-16
ArcFace	31.67%	ResNet-50
GhostFaceNet	13.33%	GhostNet (lightweight)

Critical Findings

30pp

Surrogate Performance Gap

Facenet512 (43.33%) vs GhostFaceNet (13.33%)

#1

Most Impactful Factor

Surrogate choice outweighs algorithm design

Why GhostFaceNet Fails

Its lightweight GhostNet backbone computes "ghost features" via cheap linear operations, producing gradient maps that are highly specific to its own unusual internal geometry and fail to describe universal facial identity features.

Practical Implication

In a real-world black-box attack scenario, an attacker should choose the largest, most standard surrogate model available (e.g., Facenet512, VGG-Face) for maximum transferability.

What if GhostFaceNet is Excluded?

Scenario	Overall Rate	Rank
GRA — All 4 Surrogates	31.46%	🥉 3rd
GRA — Excl. GhostFaceNet	**37.50%**	🥇 **1st**
BSR	36.46%	🥇 1st
DeCowA	32.50%	🥈 2nd
SI-NI-FGSM Baseline	29.17%	4th

Key Insight

GhostFaceNet's weak lightweight architecture single-handedly drags GRA's overall average down by 6.04 percentage points. On any reasonable surrogate (Facenet512, VGG-Face, ArcFace), GRA comfortably outperforms every other intern attack and ranks #1.

This is not a flaw in GRA — it is a fundamental limitation of using lightweight ghost-module networks as adversarial surrogates. GRA requires rich, high-dimensional gradient signal to perform its neighborhood sampling effectively.

Analysis — Why GRA Works

Gradient Diversity

VIEWS/STEP: 1 → 20

PGD uses 1 gradient view. MI-FGSM adds momentum but still uses 1 view per step. GRA averages 20 neighborhood samples per step, aggressively smoothing the surrogate's local loss landscape and destroying model-specific gradient noise.

Attack	Views/Step	Overall
PGD	1	16.46%
MI-FGSM	1	26.04%
GRA	**20**	**31.46%**

Decay Indicator as Regularizer

SIGN DECAY Γ = 0.94

GRA has a built-in anti-oscillation mechanism. Pixels whose gradient sign flips between iterations are progressively penalized (multiplied by γ=0.94 each flip), ensuring the final perturbation is dominated by stable, universally-effective signal.

Condition	Multiplier
Sign stable	×1.0
Sign flips	×0.94
After 5 flips	×0.74

Cosine Relevance Weighting

ADAPTIVE GRADIENT BLENDING

The corrected gradient is a weighted blend of the current gradient and the neighborhood average, controlled by their cosine similarity. When they agree, it trusts the current gradient. When they disagree, it leans toward the smoother neighborhood average.

cos_sim	Blend
1.0 (agree)	100% current grad
0.5 (neutral)	50/50 blend
0.0 (disagree)	100% avg grad

Limitations & Future Work

Limitations

Surrogate Dependence

GRA's performance varies wildly based on surrogate choice (13.33% to 43.33%). The algorithm cannot compensate for a poor surrogate's weak gradient signal. Ensemble surrogates could address this.

High Compute Cost

20 neighborhood samples per iteration means T=5 attack requires 100 forward/backward passes vs 5 for PGD. Significantly slower, especially on large models like VGG-Face.

Input-Space Only

GRA operates purely on input pixels and lacks intermediate feature-map disruption seen in more advanced attacks. May limit effectiveness against models with heavy input preprocessing.

Future Work

01

Ensemble Surrogate GRA

Run GRA simultaneously across all 4 surrogate models, aggregating gradients before the update step to eliminate surrogate dependence.

02

GRA + BSR Hybrid

Combine GRA's gradient relevance correction with BSR's spatial block shuffling to stack both diversity mechanisms for maximum transferability.

03

Intermediate Layer Disruption

Apply Intermediate Layer Attack (ILA) alongside GRA momentum to add feature-map level disruption on top of input-space perturbations.

Final Result Summary

Full Attack Rankings

Rank	Attack	Overall	Dodging	Impersonation
1	BSR	36.46%	49.17%	23.75%
2	DeCowA	32.50%	43.33%	21.67%
3	**GRA (Ours)**	**31.46%**	**40.00%**	**22.92%**
4	SI-NI-FGSM	29.17%	34.58%	22.08%
5	MI-FGSM	26.04%	32.92%	18.33%
6	MI-ADMIX-DI-TI	23.96%	31.25%	16.67%
7	TI-FGSM	20.00%	26.67%	13.33%
8	PGD	16.46%	20.83%	12.50%

Key Conclusions

3rd

Overall Ranking

Out of 8 evaluated transfer attacks

22.92%

Best Impersonation in Top 3

Highest among BSR, DeCowA, and GRA

Why GRA Matters

GRA proves that gradient relevance — not just diversity — is the key to transferability. By statistically filtering model-specific gradient noise via cosine similarity, it achieves competitive performance with a principled, mathematically grounded approach.

Critical Insight

Surrogate choice dominates algorithm performance. Facenet512 as surrogate achieves 43.33% while GhostFaceNet achieves only 13.33% — a 30pp gap that no algorithm can bridge.

Thank You

- Krish Bansal — DTU
- Full Implementation: **GitHub Repo**
- Reference: "Boosting Adversarial Transferability via Gradient Relevance Attack" — ICCV 2023

ICCV 2023 GRA TRANSFER ATTACKS FACE RECOGNITION